

11th International Conference on Information Technology and Quantitative Management (ITQM 2024)

A Study on Fake Review Detection Based on RoBERTa and Behavioral Features

Jinhao Liu^a, Pei Quan^{a, *}, Wen Zhang^a

^aCollege of Economics and Management, Beijing University of Technology, Beijing 100124, China

Abstract

With the rapid development of e-commerce platforms, how to better identify and filter fake reviews has become an urgent issue for the healthy development of the e-commerce industry. However, traditional fake review identification methods cannot effectively solve this problem. They are easily affected by text changes, language differences, and context, and fake reviewers may take measures to blur their behavioral characteristics, making them difficult to detect by behavior-based algorithms. Large language models can capture the contextual relationships of the entire text through self-attention mechanisms, thereby understanding the overall meaning and emotional tendency of the review, which enables them to more effectively judge the authenticity of the review. In addition, large models have strong generalization ability and a certain degree of interpretability, which also makes them suitable for research on fake review text identification. Since fake reviews have an asymmetric distribution feature, this paper uses the RoBERTa model to extract information and combine it with the behavioral features in traditional research for model training. Compared with traditional methods, the accuracy rate is improved by nearly 3%.

© 2024 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the 11th International Conference on Information Technology and Quantitative Management

Keywords: Fake review detection; Large language model; Imbalanced data;

1. Introduction

Online product reviews play a crucial role in shaping consumer purchasing decisions and influencing product sales. According to a PowerReviews survey, 95% of consumers read online reviews, and 86% of consumers believe that online reviews are important information for making purchasing decisions [1]. In 2020, 87% of consumers would read online reviews of local businesses, a significant increase from 67% in 2010 [2]. And

* Corresponding author.

E-mail address: quanpei@bjut.edu.cn.

consumers are more likely to purchase a product with 5 online reviews than a product with no reviews [3]. Moreover, the revenue of e-commerce companies is highly correlated with online consumer reviews. For example, a one-star increase in a restaurant's rating can increase revenue by 5%-9% [4], and a restaurant with an extra half-star can have over 50% more reservations [5].

However, the surge in review manipulation on e-commerce platforms has led to the prevalence of fraudulent online reviews. Despite the significant investment made by online platforms to combat fraudulent reviews, fraudulent reviews remain a critical issue affecting online shopping. A consumer survey report found that 80% of consumers reported reading fraudulent reviews, and 33% of consumers identified multiple fraudulent reviews [6]. Furthermore, despite the existence of filtering algorithms, 15%-30% of online reviews are still manipulated [7]. Amazon spends over \$500 million annually to combat fraud, including fraudulent reviewers, with over 8,000 employees working on fraud detection [8].

Simple methods based on text content struggle to capture all variations, as review content can be influenced by context, and certain words may have different meanings in different contexts. Therefore, relying solely on text content to detect fake reviews may overlook the importance of context. Additionally, fake reviews may use vocabulary similar to that of genuine reviews, making it difficult for content-based methods to distinguish accurately. Hence, for efficient detection of fake reviews, we need more sophisticated language models that also incorporate the behavioral characteristics of reviewers.

In this paper, we propose a new framework that employs the large language model RoBERTa, integrating the resulting text vectors with behavioral features, and utilizes data augmentation to address the issue of data imbalance, ultimately achieving favorable results.

2. Literature Review

Jindal and Liu [13] first proposed the fake review research problem in 2008 and found that fake reviews are prevalent. They first used duplicate review detection to detect data, then used supervised learning and a large number of manually labeled training samples for detection, and finally proved that the logistic regression model is very effective.

Fake review identification methods mainly focus on the comprehensive features, reviewer behavior and review targets of reviews to characterize the abnormal behavior of reviewers. Rayana et al. [28] proposed a new comprehensive method called SpEagle, which utilizes all metadata (text, timestamps, ratings) and relational data (networks) and utilizes them together in a unified framework to identify suspicious users, fake reviews, and products targeted by spam reviews. Kumar et al.[17] derived the distribution of key RBFs (Radial Basis Functions) to reflect the behavioral differences between real and fake reviewers. Zhang et al.[31] leverage weighted Latent Dirichlet Allocation to uncover hidden thematic patterns within reviewer behavior based on their reviews. They combine this with asymmetric KL divergence, a method specifically suited for comparing reviewers. This approach prioritizes identifying fake reviewers, the minority group, when using a nearest neighbor classification algorithm. Zhang et al.[33] developed an Expectation Maximization (EM) and Kullback – Leibler (KL) divergence based method to detect fake reviews online. It uses EM to model reviewers' topics and KL divergence to compare them, effectively spotting fraud. Overall, current methods tend to use either behavioral features or textual content. Moreover, the issue of data imbalance, where genuine reviews far outnumber fake ones in reality, has not received sufficient attention. In addition, research on large language models in the field of fake review detection is scarce. In our study, we aim to focus on overcoming and addressing these issues.

3. Methodology

Fig. 1 shows the architecture of our proposed fake review identification method, which can be roughly divided into three modules: Review Representation, Review Augmentation, Review Identification. The main components of this work are detailed as follows:

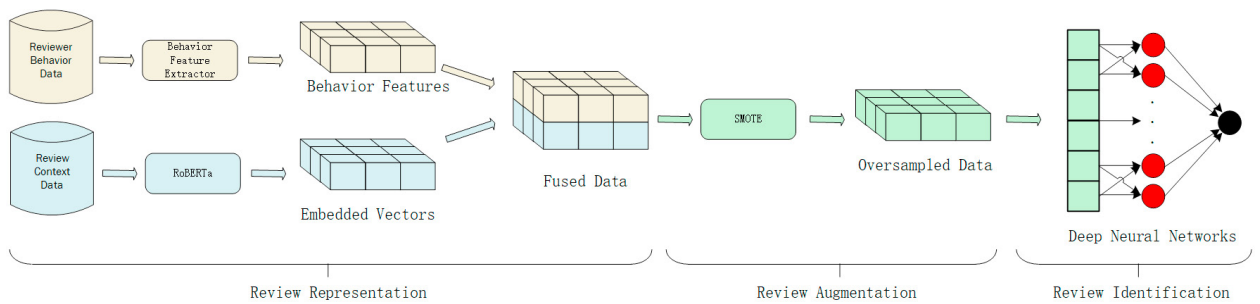


Fig. 1. The framework of proposed fake review identification method

3.1. Review Text Embedding Generation

To convert text into data that can be processed by computers, we need to convert the text into vector form. Word embedding is a technique that maps high-dimensional data to low-dimensional space. Compared with traditional embedding methods, the RoBERTa model also includes an additional step called attention mechanism. The attention mechanism allows the model to focus on the most important parts of the text. This allows Roberta to better capture the context and semantic information of the text, thereby generating more representative text vectors. Many researchers believe that linguistic clues such as review length, subjectivity of review content, number of nouns, verbs and adjectives, writing style (e.g. using specific types of words), etc., can be used to accurately distinguish between real and fake reviews, and these clues can all be captured by the vectors obtained through word embedding.

3.2. Behavioral Features Extraction

In designing solutions, we consider the potential association of these features with fake reviews and extract relevant data to train our detection models. By analyzing these behavioral features, we can better understand and identify potential fake review behaviors. Ultimately, a method that combines text analysis with behavioral feature analysis may provide a more comprehensive and accurate fake review detection system. The behavioral features used in this experiment include:

- Activity Window (AW): This calculates the timestamp difference between the author's first and last reviews. A short activity window may indicate rapid consecutive reviews, which could be related to fake reviews.
- Number of Positive Ratings (PR): Represents the number of positive ratings each author has received. Fake reviewers might receive an abnormal number of positive ratings, especially if these ratings appear to be self-generated.
- Number of Negative Ratings (NR): Represents the number of negative ratings each author has received. Fake reviewers might attempt to damage a competitor's reputation through negative ratings.
- Review Order (OOR): This feature represents the ranking order of reviews or ratings for each product over time. For example, a new product suddenly receiving a large number of positive reviews could be an indicator of fake reviews.

These features cover multiple behavioral characteristics of fake reviewers and can reflect different behavioral patterns of fake reviewers. For example, fake reviewers often post a large number of reviews and a large number of reviews in a short period of time to quickly improve the credibility of their accounts. They may also manipulate product ratings by posting fake positive or negative ratings. Review order can also be used to identify fake reviews, as fake reviewers may post a large number of reviews in a short period of time, and the order of these reviews may be unnatural.

3.3. Review Augmentation

Since the experimental data has a highly unbalanced feature, this experiment uses the SMOTE data augmentation algorithm. SMOTE [21] (Synthetic Minority Over-sampling Technique) is a commonly used data augmentation method used to solve the data imbalance problem in machine learning. The basic idea of SMOTE is to generate new minority class samples by synthesizing minority class samples, thereby increasing the number of minority class samples and making the number of samples in the training set more balanced.

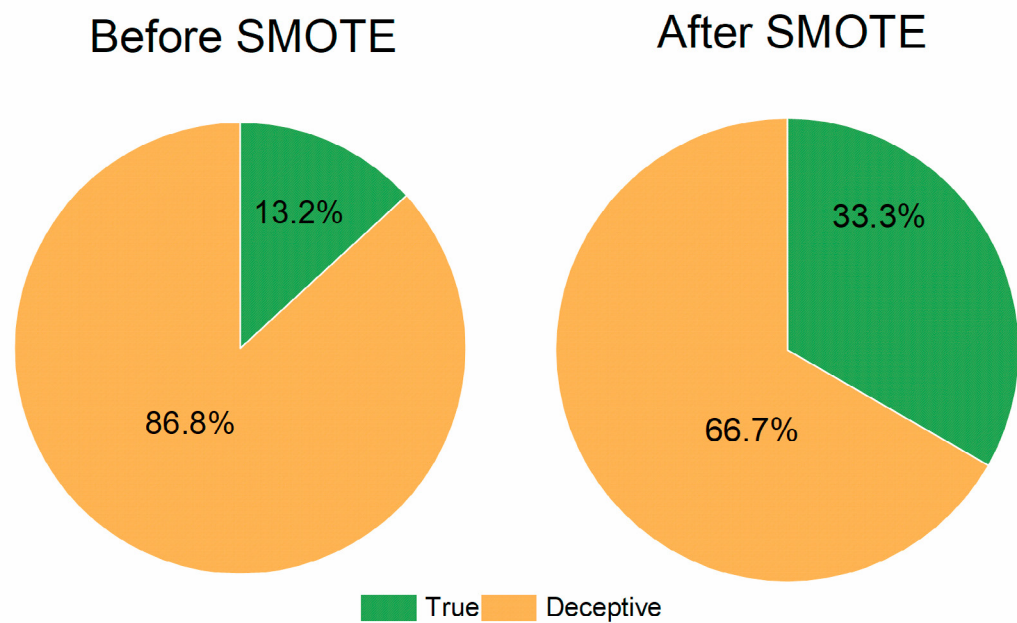


Fig. 2 Distribution of True and False Samples After SMOTE Data Augmentation

As can be seen from Fig. 2, after applying SMOTE technology, the sample imbalance problem in the review data has been greatly improved, which will ensure that the model is more fair during the training process.

4. Experimental Results

4.1. Experimental Data

The dataset used in the experiment is a collection of restaurant reviews from Yelp.com, shared by Rayana and Akoglu [28]. It is widely used in research on fraudulent review detection and fraudulent reviewer detection. The dataset contains 608,598 reviews of 5,044 restaurants posted by 262,277 reviewers. Of the 262,277 reviewers, 62,228 were flagged as fraudsters. In addition, the dataset also contains information about each review, including star rating, review content, review date, and target restaurant.

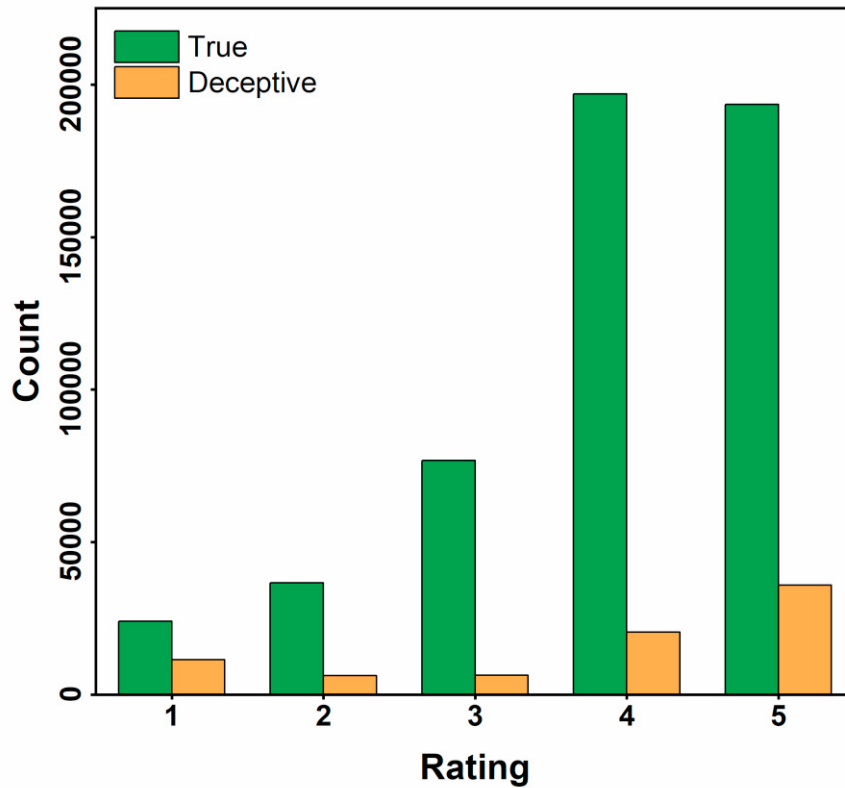


Fig. 3 Distribution of Star Ratings and True/Deceptive Numbers

Fig. 3 shows the True/Deceptive (False) distribution of different star ratings in the fraudulent review dataset. As can be seen from the figure, the number of 5-star reviews for true reviews exceeds 175,000, far exceeding the number of deceptive reviews. The number of false reviews is also relatively large, but still far less than the number of true reviews. The number of 1-star, 2-star, and 3-star reviews is relatively small, whether they are true or false reviews. The number of 4 and 5 star reviews is very high, with true reviews dominating.

4.2. Performance Analysis:

To combine the advantages of deep learning methods and behavioral feature recognition methods, we added behavioral features to the training model, and the training results are as follows:

Table 1: Results of Different Models

Model	Precision	Recall	F1
Naive Bayes Classifier	78.98%	61.43%	70.20%
Linear Support Vector Machine	71.19%	66.77%	68.98%
Random Forest	66.79%	64.94%	65.86%
k-Nearest Neighbors Classifier (k=250)	65.87%	63.26%	64.57%
RoBERTa+LSTM+BFs	85.94%	86.54%	85.32%
RoBERTa+CNN+BFs	84.73%	83.82%	84.28%
RoBERTa+LSTM-CNN+BFs	87.65%	89.87%	88.76%

As can be seen from the table 1, although traditional machine learning algorithms can make effective judgments, their overall accuracy is poor and it is difficult to achieve the desired results. Compared to traditional

machine learning methods, our model achieves significantly better results after using deep neural networks. One possible reason is that the extracted features have complex interconnections, which traditional machine learning methods struggle to capture.

4.3. Model Analysis

Our CNN model starts with a 1D convolutional layer with a kernel size of 5. The activation function used here is ReLU. The second layer is a global max pooling layer. It takes the maximum value from each feature map (output of the convolutional layer) across all time steps. This reduces the sequence length to a fixed size. The third layer is a dense layer with 64 units and a ReLU activation function. It learns higher-level features by combining information from the previous layers. The final layer is the output layer with a single unit and a sigmoid activation function.

Our LSTM network has two layers. The first LSTM layer is bidirectional and returns sequences. The second LSTM layer is also bidirectional but with 64 units. It does not return sequences, meaning it aggregates information across the entire sequence. A dense layer (attention) with a single output unit and a hyperbolic tangent (tanh) activation function is applied to the output of LSTM layer 2. The result is flattened using a flatten layer. An activation layer with softmax activation is applied to obtain attention weights.

The first layer of our combined model is a 1D convolutional layer with a kernel size of 3. This is followed by a flatten layer to convert the convolutional output into a 1D vector. Next, an activation layer with ReLU activation is applied to the flattened output. In parallel, we process the data with two bidirectional LSTM layers with 64 units each, both returning sequences. We then add another attention layer using MultiHeadSelfAttention to the output. This is followed by a second flatten layer for the attention output. A concatenation layer combines the flattened convolutional output and the attention output. We then add a dropout layer with a dropout rate of 0.2. This is followed by a dense layer with 64 units and ReLU activation. Finally, we use the final output layer with a single unit and sigmoid activation for binary classification.

Table 2: Results of different ablation studies

Model	Precision	Recall	F1
Glove+LSTM+BFs	82.21%	83.77%	82.90%
Glove+CNN+BFs	81.56%	80.87%	81.20%
Glove+CNN-LSTM+BFs	84.20%	82.97%	83.58%
RoBERTa+LSTM	84.43%	85.71%	85.07%
RoBERTa+CNN	83.56%	82.87%	83.22%
RoBERTa+CNN-LSTM	85.20%	86.21%	85.71%
RoBERTa+LSTM+BFs(Without SMOTE)	78.43%	69.64%	74.03%
RoBERTa+CNN+BFs(Without SMOTE)	76.32%	70.35%	73.34%
RoBERTa+CNN-LSTM+BFs(Without SMOTE)	79.65%	71.01%	75.33%
RoBERTa+LSTM+BFs	85.94%	86.54%	85.32%
RoBERTa+CNN+BFs	84.73%	83.82%	84.28%
RoBERTa+LSTM-CNN+BFs	87.65%	89.87%	88.76%

As shown in the table above, models using RoBERTa generally outperform those using GloVe in terms of Precision, Recall, and F1 scores. This indicates that RoBERTa's pre-trained representation capability is stronger. Without the use of SMOTE data augmentation, the performance of the RoBERTa model significantly declines, especially in Recall, which may suggest that SMOTE is effective for dealing with imbalanced datasets. The combined CNN-LSTM model structure seems to achieve better results in both GloVe and RoBERTa versions,

which might be due to CNN's ability to capture local features, while LSTM can capture long-term dependencies. The RoBERTa+LSTM-CNN+BFs model performs best across all metrics, with an F1 score reaching 88.76%, demonstrating the advantages of the model structure and feature engineering. Generally, the models perform better when including user behavior features, indicating that user behavior data is valuable for fake review detection. Overall, these results demonstrate the performance enhancement of fake review detection tasks due to RoBERTa, feature engineering, and data augmentation techniques.

5. Conclusion

In this work, we propose a fake review identification framework with a two-fold contribution. Firstly, it combines the strengths of large language models with traditional machine learning approaches. By incorporating user behavior features alongside text information extracted by RoBERTa, the model gains a deeper understanding of users and reviews. This approach enhances the model's generalization ability and reduces training costs. Secondly, it improves model robustness. Behavioral features reflect users' genuine actions, further strengthening the model's resilience.

In the future, we can employ more effective behavioral features for improved model performance. By collecting more high-quality behavioral feature data, we can enhance the model's generalizability. Grouping similar samples will help us gain a better understanding of data distribution and structure, ultimately improving generalization ability. Selecting the most representative features will reduce feature space dimensionality and improve model performance. Overall, the proposed method presents a promising approach for detecting fraudulent online reviewers. It has the potential to significantly improve the accuracy and efficiency of online review manipulation detection.

Acknowledgements

This work was supported in part by Beijing Natural Science Foundation (Grant No. 9244021, No. 9222001); the National Natural Science Foundation of China under Grant Nos. 72174018 and 71932002; R&D Program of Beijing Municipal Education Commission (SM202410005007); Beijing Postdoctoral Research Foundation (Grant No. 2023-zz-176); the Philosophy and Sociology Science Fund from Beijing Municipal Education Commission (SZ2021110005001).

References

- [1] E.C. Malthouse, G. Askalidis, How Do Rating and Reviews Affect Conversion Rates on E-Commerce Sites?, n.d. <https://spiegel.medill.northwestern.edu/from-reviews-to-revenue/>.
- [2] Rosie Murphy, Local Consumer Review Survey 2020, 2020. <https://www.brightlocal.com/research/local-consumer-review-survey/>.
- [3] T. Collinger, How Online Reviews Influence Sales, 2017. <https://spiegel.medill.northwestern.edu/online-reviews/>.
- [4] M. Luca, Reviews, Reputation, and Revenue: The Case of Yelp.Com, 2012.
- [5] M. Anderson, J. Magruder, Learning from the Crowd: Regression Discontinuity Estimates of the Effects of an Online Review Database, *Econ. J.* 122 (2012) 957–989.
- [6] Zhang, Y., & Chen, Q. (2018). Detecting Fake Reviews Using Deep Learning. arXiv preprint arXiv:1810.02602. :
- [7] M. Luca, G. Zervas, Fake it till you make it: Reputation, competition, and yelp review fraud, *Manage. Sci.* 62 (2016) 3412–3427.
- [8] W. Wang, D. Sun, The improved AdaBoost algorithms for imbalanced data classification, *Inf. Sci. (Ny)*. 563 (2021) 358–374.
- [9] J. Boyce, It is time to kick counterfeit goods off Amazon, Facebook Marketplace and other e-commerce sites, (n.d.). <https://www.marketwatch.com/story/it-is-time-to-kick-counterfeit-goods-off-amazon-facebook-marketplace-and-other-e-commerce-sites-11617975097>.
- [10] W. Zhang, Q. Wang, X. Li, T. Yoshida, J. Li, DCWord: A Novel Deep Learning Approach to Deceptive Review Identification by Word Vectors, *J. Syst. Sci. Syst. Eng.* 28 (2019) 731–746.
- [11] Y.Liu et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach," arXiv:1907.11692(2019)
- [12] W. Zhang, Y. Du, T. Yoshida, Q. Wang, DRI-RCNN: An approach to deceptive review identification using recurrent convolutional neural network, *Inf. Process. Manag.* 54 (2018) 576–592.

- [13] Jindal N, Liu B. Opinion spam and analysis[C]//Proceedings of the 2008 international conference on web search and data mining. 2008: 219–230.
- [14] D. Plotkina, A. Munzel, J. Pallud, Illusions of truth—Experimental insights into human and algorithmic detections of fake online reviews, *J. Bus. Res.* 109 (2020) 511–523.
- [15] H. Zhang, M. Li, RWO-sampling: a random walk over-sampling approach to imbalanced data classification, *Inform. Fusion* 20 (2014) 99–116.
- [16] B. Manaskasemsak, J. Tantisuwankul, A. Rungsawang, Fake review and reviewer detection through behavioral graph partitioning integrating deep neural network, *Neural Comput. Appl.* (2021).
- [17] N. Kumar, D. Venugopal, L. Qiu, S. Kumar, Detecting Review Manipulation on Online Platforms with Hierarchical Supervised Learning, *J. Manag. Inf. Syst.* 35 (2018) 350–380.
- [18] N. Kumar, D. Venugopal, L. Qiu, S. Kumar, Detecting Anomalous Online Reviewers: An Unsupervised Approach Using Mixture Models, *J. Manag. Inf. Syst.* 36 (2019) 1313–1346.
- [19] Z. Wang, S. Gu, X. Xu, GSLDA: LDA-based group spamming detection in product reviews, *Appl. Intell.* 48 (2018) 3094–3107.
- [20] D. Zhang, L. Zhou, J.L. Kehoe, I.Y. Kilic, What online reviewer behaviors really matter? Effects of verbal and nonverbal behaviors on detection of fake online reviews, *J. Manag. Inf. Syst.* 33 (2016) 456–481.
- [21] M. Galar, A. Fernández, E. Barrenechea, F. Herrera, EUSBoost: enhancing ensembles for highly imbalanced data-sets by evolutionary undersampling, *Pattern Recogn.* 46 (2013) 3460–3471.
- [22] Nitesh V. Chawla, et al., SMOTE, *J. Artif. Intell. Res.* 16 (2002) 321–357.
- [23] B. Hooi, K. Shin, H.A. Song, A. Beutel, N. Shah, C. Faloutsos, Graph-based fraud detection in the face of camouflage, *ACM Trans. Knowl. Discov. Data.* 11 (2017).
- [24] E. Kauffmann, J. Peral, D. Gil, A. Ferrández, R. Sellers, H. Mora, A framework for big data analytics in commercial social networks: A case study on sentiment analysis and fake review detection for marketing decision-making, *Ind. Mark. Manag.* (2019).
- [25] R. Bhargava, A. Baoni, Y. Sharma, Composite sequential modeling for identifying fake reviews, *J. Intell. Syst.* 28 (2019) 409–422.
- [26] H.A. Khorshidi, U. Aickelin, Constructing classifiers for imbalanced data using diversity optimisation, *Inf. Sci. (Ny)*. 565 (2021) 1–16.
- [27] H. Kaur, H.S. Pannu, A.K. Malhi, A systematic review on imbalanced data challenges in machine learning: Applications and solutions, *ACM Comput. Surv.* 52 (2019).
- [28] S. Rayana, L. Akoglu, Collective opinion spam detection: Bridging review networks and metadata, in: *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2015: pp. 985–994.
- [29] Pennington, J., Socher, R., & Manning, C. D. (2014). GloVe: Global Vectors for Word Representation. *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- [30] Q.Wang, W.Zhang, J.Li, Z.Ma, Complements or confounders? A study of effects of target and non-target features on online fraudulent reviewer detection, *Journal of Business Research*, Volume 167, 2023, 114200 SSCI
- [31] W.Zhang, R.Xie, Q.Wang, Y.Yang, J.Li..A novel approach for fraudulent reviewer detection based on weighted topic modelling and nearest neighbors with asymmetric Kullback–Leibler divergence[J]. *Decision Support Systems*, 2022, 157: 113765.
- [32] W. Zhang, Q. Wang, X. Li, T. Yoshida, J. Li.: DCWord: A Novel Deep Learning Approach to Deceptive Review Identification by Word Vectors. *Journal of Systems Science and Systems Engineering*, 28(6):731-746, December 2019.
- [33] W.Zhang, G.Qin, and Q.Wang. 2022. Handling Imbalance in Fraudulent Reviewer Detection based on Expectation Maximization and KL Divergence. In *IEEE/WIC/ACM International Conference on Web Intelligence and Intelligent Agent Technology (WI-IAT '21)*. Association for Computing Machinery, New York, NY, USA, 421–427. <https://doi.org/10.1145/3498851.3498989>.